SIMATS ENGINEERING
CO2 ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Duration : 1 Hour Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

I Arunprasath R (192524077) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Arunprasath R

Annexure A

Scenario Based Assignment

1. AI Drones for Flood Relief (Greedy Best-First vs A*)

Scenario: A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic: roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to a heuristic estimate (straight-line distance), partial real-time updates about blocked paths and variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty.

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic. iii. Analyze the impact of heuristic accuracy on both algorithms.

iv. Discuss how dynamic changes (blocked paths) affect search performance.

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety.

ANS:

i. Greedy Best-First Search in Flood Relief

Greedy Best-First Search (GBFS) selects the path that appears closest to the destination based on a heuristic function.

- **Working:** The drone uses straight-line distance to select the next location that seems closest to the medicine delivery point.
- **Strengths:**
    1. Explores fewer locations in many cases.
    2. Makes quick decisions, which is useful during emergencies.
    3. Simple to implement and understand.
- **Weaknesses:**
    1. May select a shorter route with dangerous weather conditions.
    2. Does not consider the total travel cost.
    3. May choose a blocked path and require replanning.
- **Example:** The drone selects a route that appears closest to the hospital, but the route passes through a region with strong winds.

Conclusion: GBFS is fast but may not find the safest or lowest-cost route.

ii. A* Algorithm in Flood Relief

A* selects a path by considering both the actual cost travelled and the estimated cost remaining to the destination.

It uses the evaluation function:

$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$= Actual cost from the starting point to the current location.
- $h(n)$= Estimated cost from the current location to the destination.
- $f(n)$= Estimated total cost of the path.

Working:

1. The drone identifies possible routes to the destination.
2. It calculates the actual travel cost, including terrain and weather conditions.

3. It estimates the remaining distance using a heuristic.
4. It selects the path with the lowest estimated total cost.
5. If a path becomes blocked, the drone updates the route and searches again.

Advantages:

- Considers both current travel cost and estimated remaining cost.
- Can find an optimal path when the cost model and heuristic satisfy the required conditions.
- Can avoid routes with high weather-related costs when those costs are correctly represented.

Example: Although one route is shorter, it has strong winds and higher travel costs. A* may select a slightly longer route with lower overall cost and risk.

Conclusion: A* generally provides a better balance between search efficiency and route quality.

iii. Impact of Heuristic Accuracy on Both Algorithms

A heuristic estimates the remaining cost from the current location to the destination.

Factor	**Greedy Best-First Search**	**A* Algorithm**
Heuristic function	Uses $h(n)$	Uses $g(n) + h(n)$
Accurate heuristic	Usually helps reach the destination faster	Usually improves search efficiency
Inaccurate heuristic	May select a poor route	May explore unnecessary locations or select a poor route
Optimality	Not guaranteed	Guaranteed under appropriate conditions
Flood conditions	May underestimate dangerous routes	Can account for dangerous routes if their costs are modelled

Example: Straight-line distance may suggest that a destination is nearby, but strong winds or a flooded region may make the route expensive or unsafe.

Therefore, the drone should use updated information about weather, terrain, and blocked paths to improve route selection.

iv. Impact of Dynamic Changes (Blocked Paths)

Flood conditions can change suddenly, making previously available routes inaccessible.

Greedy Best-First Search:

1. May continue toward a blocked route because it appears closest to the destination.
2. Must update its search when the blockage is detected.
3. May find an alternative quickly, but the new route may have a higher travel cost.

*A Algorithm:**

1. Updates the costs and availability of affected routes.
2. Searches for an alternative route using actual and estimated costs.
3. Can identify a lower-cost alternative when sufficient information is available.
4. May require additional computation when many routes change.

Example: If a strong wind makes the original route unsafe, the drone marks that route as unavailable and searches for another safe route.

Important point: Standard GBFS and A* do not automatically adapt to every environmental change. They need updated information and may need to run the search again. Frequent changes can increase computation time.

v. Recommendation and Justification

Recommended algorithm: A* with real-time route updates.

Criteria	**Greedy Best-First Search**	**A* Algorithm**
Search speed	Often fast	May require more computation
Travel cost	Does not consider total path cost	Considers total path cost
Optimality	Not guaranteed	Guaranteed with suitable cost assumptions and an admissible heuristic
Safety	Depends heavily on additional safety	Can account for safety through route costs and constraints

	checks	
Dynamic changes	Requires replanning when necessary	Requires replanning or a suitable dynamic-search extension

Justification

1. **Minimum travel cost:** A* considers both the cost already incurred and the estimated remaining cost.
2. **Improved safety:** Weather, terrain, and risk can be incorporated into the route cost, while dangerous routes can be prohibited.
3. **Dynamic adaptation:** Real-time updates allow the drone to reconsider its route when a path becomes blocked.
4. **Better decision-making:** A* avoids relying solely on straight-line distance when choosing a route.
5. **Optimality:** A* can find an optimal route when its cost model is accurate enough and the heuristic satisfies the required conditions.

However, A* is not automatically safer or faster in every situation. The drone must receive reliable updates, account for uncertainty, and satisfy safety constraints.

For environments that change frequently, *A combined with a dynamic replanning algorithm such as D* Lite** can be considered.

12. Mobile Tower Frequency Assignment (Graph Colouring CSP)

Scenario: A telecom operator must assign one of three frequency bands to each of its mobile towers in a district. Towers whose coverage areas overlap must use different bands to avoid interference. Some towers in remote valleys do not overlap with any others.

Tasks:

i. Model the problem as a graph colouring CSP and draw a constraint graph for a small example (at least 6 towers).

ii. Explain the MRV, degree and least-constraining-value heuristics, and apply them to your example. iii. Show how forward checking detects a failure early, using your example.

iv. Explain how independent subproblems and tree-structured constraint graphs make the problem easier. v. Discuss what changes if a fourth frequency band becomes available

ANS:

i. Model the Problem as a Graph Colouring CSP

A **Constraint Satisfaction Problem (CSP)** consists of variables, domains, and constraints. In mobile tower frequency assignment, each tower must receive a frequency band without interfering with neighbouring towers.

CSP Model

- **Variables:** Six mobile towers — A, B, C, D, E and F.
- **Domains:** Three frequency bands: Red (R), Green (G) and Blue (B).
- **Constraints:** Towers whose coverage areas overlap must use different frequency bands.
- **Goal:** Assign a frequency to every tower without violating any constraints.

Constraint Graph

Consider the following example:

- A overlaps with B and C.
- B overlaps with A, C and D.
- C overlaps with A, B and D.
- D overlaps with B, C and E.
- E overlaps with D.
- F does not overlap with any tower.

Graph representation:



\ /
D
|
E

F

Here, each node represents a tower, and each edge represents an interference constraint.

CSP representation:

- Variables: X = {A, B, C, D, E, F}
- Domains: D(A) = D(B) = D(C) = D(D) = D(E) = D(F) = {R, G, B}
- Constraints: A ≠ B, A ≠ C, B ≠ C, B ≠ D, C ≠ D and D ≠ E.

Tower F is independent because it has no overlapping coverage area.

One Valid Frequency Assignment

Tower	Frequency band
A	Red
B	Green
C	Blue
D	Red
E	Green
F	Red

This assignment satisfies all interference constraints.

ii. MRV, Degree and Least-Constraining-Value Heuristics

These heuristics help a CSP solver choose variables and values efficiently.

1. Minimum Remaining Values (MRV)

Definition: Select the unassigned variable with the fewest legal values remaining in its domain.

Application:

Initially, all towers have three possible frequency bands. Suppose A is assigned Red.

- B cannot use Red.
- C cannot use Red.
- D, E and F still have three available bands.

Now B and C each have two remaining choices: Green or Blue.

MRV selects either B or C because they have fewer remaining values than the other unassigned towers.

Advantage: Detects difficult variables early and reduces unnecessary search.

2. Degree Heuristic

Definition: Select the unassigned variable that has the largest number of constraints with other unassigned variables.

Application:

In our example:

Tower	Degree (number of neighbours)
A	2
B	3
C	3
D	3
E	1
F	0

Initially, B, C and D have the highest degree of 3.

If MRV produces a tie, the degree heuristic can select B or C because they constrain more neighbouring towers.

Advantage: Assigning highly connected towers early can prevent future conflicts.

3. Least-Constraining-Value (LCV)

Definition: Choose the value that eliminates the fewest possible values from neighbouring unassigned variables.

Application:

Suppose A is assigned Red.

For B, the available values are Green and Blue.

- Assigning Green to B removes Green from C and D.
- Assigning Blue to B removes Blue from C and D.

Both values have the same effect in this example. Therefore, either can be selected.

In a more complex network, LCV prefers the frequency that leaves the most choices available for neighbouring towers.

Advantage: Preserves flexibility for future assignments and reduces backtracking.

Summary

Heuristic	Purpose	Selection rule
MRV	Select a variable	Fewest legal values
Degree	Break variable-selection ties	Most unassigned neighbours
LCV	Select a value	Eliminates the fewest choices for neighbours

iii. How Forward Checking Detects Failure Early

Forward checking updates the domains of neighbouring unassigned variables after assigning a frequency to a tower.

If any unassigned tower has an empty domain, the algorithm detects failure immediately and backtracks.

Example

Consider towers A, B and C, where every tower overlaps with the other two.

Each tower has the domain:

$$D(A) = D(B) = D(C) = \{R, G, B\}$$

Suppose the solver makes these assignments:

Step 1: Assign A = Red.

The domains become:

- B = {Green, Blue}
- C = {Green, Blue}

Step 2: Assign B = Green.

Since B overlaps with C, Green is removed from C's domain.

Therefore:

$$D(C) = \{Blue\}$$

Step 3: Assign C = Blue.

All constraints are satisfied.

Failure Detection Example

Now consider four mutually overlapping towers A, B, C and D, with only three frequency bands.

1. Assign A = Red.
2. Assign B = Green.
3. Assign C = Blue.
4. Forward checking examines D.

Since D overlaps with A, B and C, it cannot use Red, Green or Blue.

Thus:

$$D(D) = \emptyset$$

Result: Forward checking detects failure before attempting to assign D. The solver must backtrack and reconsider an earlier assignment.
This example also demonstrates that four mutually interfering towers cannot be assigned only three frequency bands.
Advantage: Forward checking identifies conflicts early and avoids exploring impossible assignments unnecessarily.

iv. Independent Subproblems and Tree-Structured Constraint Graphs

1. Independent Subproblems

An independent subproblem is a group of variables that has no constraints connecting it to another group.
In our example, tower F does not overlap with any other tower.
Therefore, F can be assigned any frequency independently.
For instance:

- Main network: A, B, C, D and E.
- Independent tower: F.

The solver can solve the main network first and then assign a frequency to F.
Benefits:

- Reduces the size of the problem being solved at one time.
- Allows independent groups to be solved separately.
- Can improve efficiency, especially when the network contains many disconnected groups.

2. Tree-Structured Constraint Graphs

A tree-structured constraint graph is a connected graph with no cycles.
For example:
A — B — C — D — E
Each tower is connected through a single chain, with no cycles.
With three frequency bands, a solution can be found by assigning a frequency to A and then choosing a different frequency for each successive tower.
For example:

Tower	Frequency
A	Red
B	Green
C	Red
D	Green
E	Red

Every pair of adjacent towers has different frequencies.
Benefits:

- Avoids many complications caused by cycles.
- Can be solved efficiently using constraint propagation and variable ordering.
- For a tree-structured binary CSP, a solution can be found in time proportional to the number of variables and domain size, under the standard algorithm.

Conclusion: Independent components and tree structures simplify frequency assignment by reducing conflicts and search effort.

v. What Changes if a Fourth Frequency Band Becomes Available?

Suppose a fourth frequency band, Yellow (Y), is introduced.
The domain of every tower becomes:

$$D(X) = \{R, G, B, Y\}$$

Changes in the CSP

1. Larger domains

Each tower now has four possible frequency bands instead of three.

2. Fewer conflicts

More choices are available when neighbouring towers require different frequencies.

3. Easier problem solving

Some assignments that were impossible with three bands may become possible with four.

Example: Consider four mutually overlapping towers A, B, C and D.

With three frequency bands, no valid assignment exists because all four towers must have different bands.

With four frequency bands, a valid assignment is:

Tower	Frequency band
A	Red
B	Green
C	Blue
D	Yellow

All four towers have different frequencies, so interference constraints are satisfied.

4. Reduced backtracking

The additional band may reduce domain wipeouts and the need to reconsider earlier assignments.

5. Increased resource usage

An additional frequency band may require more spectrum resources and could increase operational costs or regulatory complexity.

Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem